

Deep Learning for Site Seismic Amplification: A Regression Approach using KiK-net Data

Sofia De Angelis, Matteo Pinato.

Summary

One of the key factors influencing seismic risk is the accurate assessment of local seismic response. Traditional empirical methods, often based on multiple linear regression (MLR) models, exhibit significant limitations in capturing the complex nonlinear interactions between soil properties and seismic wave propagation. This study proposes a **Deep Neural Network (DNN)**-based approach for predicting the seismic amplification factor, formulating the problem as a regression task aimed at quantifying site response in continuous rather than categorical terms. The model is trained using the **KiK-net database** (Japan), selected for its vertical array configuration, which enables the direct and accurate computation of empirical site transfer functions. Input variables include key geotechnical parameters and local stratigraphic characteristics. Despite computational challenges associated with encoding variable stratigraphies into fixed-size architectures, the proposed approach aims to surpass the predictive capabilities of classical linear models. Preliminary results suggest that the use of DNNs represents a promising methodology for rapid seismic microzonation and preliminary risk assessment in complex geological settings.

1 Location & Dataset

1.1 KiK-net Introduction

The *Kiban Kyoshin Network (KiK-net)*, operated by the *National Research Institute for Earth Science and Disaster Resilience (NIED[1])* of Japan, represents one of the densest and most advanced strong-motion seismic monitoring infrastructures in the world.

Established following the disastrous 1995 Kobe earthquake, KiK-net was designed with the primary goal of mitigating seismic risk through a detailed understanding of subsurface wave propagation and site effects.

The network comprises nearly 700 stations uniformly distributed across the Japanese archipelago, providing a database of exceptionally high-quality accelerometric recordings, essential for research in earthquake engineering and seismology. Each station

is also supported by detailed geotechnical information, including stratigraphic columns and shear wave velocity (V_s) profiles, obtained through downhole logging.

1.2 Sites Architecture

The distinctive feature that makes the KiK-net database unique lies in the instrumental configuration of its stations. Unlike traditional seismic networks that primarily utilize surface sensors, each KiK-net site is *designed as a vertical array* composed of a pair of co-located three-component accelerometers:

- **Surface Sensor:** Installed at ground level (GL-0m) inside a dedicated instrument shelter.
- **Borehole Sensor:** Installed at depth, directly above the surface sensor. The installation depth varies depending on local conditions, but is generally designed to reach and penetrate the so-called "engineering bedrock" (typically layers with $V_s > 700m/s$), positioning itself beneath the soft surface deposits. The average depth is often over 100 meters.

Figure 1 schematically illustrates the typical architecture of a KiK-net station, highlighting the paired arrangement of the sensors with respect to the local stratigraphy.

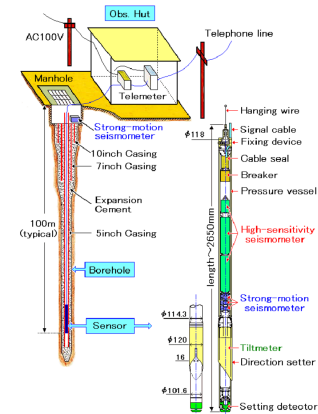


Figure 1: Configuration of the KiK-net station

1.3 Why KiK-net?

The simultaneous availability of surface and deep recordings at the same site offers a crucial methodological advantage for estimating site amplification.

In traditional methods, site effects are often estimated using techniques such as the single-sensor **horizontal-to-vertical (H/V) spectral ratio** or **the reference site method**, where the motion of a distant rock outcrop station is used as a proxy for the input motion at the base of the soil column.

Both methods, while useful, introduce uncertainties: the H/V is an approximation of the transfer function, while the distant reference site may not accurately reflect the seismic input beneath the site of interest due to spatial variability in motion.

The KiK-net configuration overcomes these limitations.

The borehole sensor, being installed in the bedrock and below the surface layers responsible for most of the amplification and nonlinearity phenomena, records a motion that can be considered a reliable approximation of the seismic input (incident motion plus downward reflected motion at the bedrock-soil interface) that excites the overlying soil column. Consequently, the site amplification can be directly calculated as the *Empirical Transfer Function* (ETF), obtained through the spectral ratio between the surface recordings (output) and the depth recordings (input).

$$ETF(f) = \frac{S_{surface}(f)}{S_{depth}(f)}$$

Where $S(f)$ represents the Fourier (or response) spectrum of the motion. This direct approach allows to:

- Effectively isolate the response of the layered soil column from source and path effects, which are common to both sensors.
- Precisely quantify the real amplification as a function of the frequency.
- To study ground nonlinearity phenomena by comparing ETFs obtained from weak seismic events (linear behavior) with those obtained from strong ground motions (nonlinear behavior), where a reduction in amplification and a shift of peaks towards lower frequencies are typically observed.

The geographical area considered for this study, characterized by a high density of KiK-net stations, is shown in *Figure 2*.

1.4 Downloaded Datasets

To construct the training dataset for the Deep Neural Network (DNN), we integrated site characterization

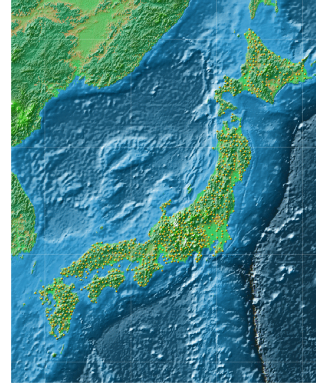


Figure 2: Geographical area of the sites

data with strong-motion records using three primary source files:

- **SiteDatabaseofK-NETandKiK-net_Strong-MotionStations_v1.0.0_20201201:** Sourced from Zhu et al. [4], this database provides comprehensive site characterization parameters for 1,742 recording stations across the K-NET and KiK-net networks in Japan. It is used to define the geological and geophysical properties of the sites in the model.
- **2025-001Loviknes-et-al19972024kikMETA.csv:** This file contains the essential metadata for the KiK-net ground-motion records (Loviknes et al., [3]), including event information (e.g., magnitude, hypocentral location) and station coordinates, serving as the linkage between source, path, and site terms.
- **2025-001Loviknes-et-al19972024kikSAB.csv:** Also derived from the dataset by Loviknes et al. [3], this file provides the ROTD50 pseudo-spectral amplitudes of the ground motions recorded at the borehole level. These values represent the input motion (bedrock condition) utilized during the DNN training phase.

The corresponding surface recording file (2025-001Loviknes-et-al19972024kikSA.csv) was strictly excluded from the input features to prevent data leakage, that occurs when information from the target variable—unavailable at the time of prediction—is inadvertently included in the training set. In this study, using surface motion as an input to predict surface response would allow the model to memorize the target rather than learning the physical transfer function from borehole to surface, thereby invalidating the model’s predictive capability on unseen data.

Data from these files were merged based on station and event identifiers to create the final input vectors for the neural network; to homogenize the feature distributions and ensure consistent numerical ranges,

a base-10 logarithmic transformation was applied to both the input and target variables following the dataset merging process.

2 Model Description

To model the non-linear relationship between seismic/site parameters and the amplification factor, we designed and implemented a **Deep Neural Network (DNN)** based on a fully connected feed-forward architecture (*Multi-Layer Perceptron*).

The primary objective of the model is to approximate the function mapping the 18 input features ($\mathbf{x}$) to the logarithmic amplification ratio (Y).

The network topology, *figure:3*, was empirically tuned to balance model complexity and generalization capability and the architecture consists of an input layer matching the feature dimensionality, so we have 18 input layers, followed by four hidden dense layers with decreasing width: $128 \rightarrow 256 \rightarrow 128 \rightarrow 64$ neurons, and a final output layer for scalar regression, so 1 neuron (linear activation) to predict the continuous target variable.

The final configuration is the result of an iterative process through 20 epochs. We observed that deeper networks did not significantly improve accuracy but increased the risk of overfitting, leading us to prefer a "funnel" architecture that progressively compresses the information.

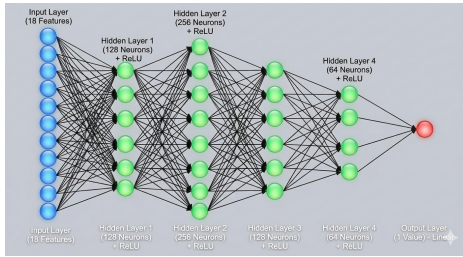


Figure 3: Deep Neural Network (DNN) Architecture with: 18 input features, 1 output (PGA Ratio) and 4 Hidden Layers with ReLU activation functions. [2]

To introduce non-linearity, the **Rectified Linear Unit (ReLU)** activation function, defined as $f(x) = \max(0, x)$, is applied after each hidden layer. This allows the network to capture complex, high-dimensional patterns in the site response data that linear models cannot resolve.

2.1 Training Strategy

To train the model it was necessary to clean the comprehensive dataset which initially comprised 434,898 seismic events spanning the period from 1997 to 2024. To ensure the reliability of the input data, a quality

control filter was applied based on the signal energy ratio (**energy_ratioSignal_B**), removing events with non-optimal signal characteristics; this step reduced the dataset to 354,717 events. Subsequently, rows containing missing values were eliminated to guarantee data completeness, resulting in a final refined dataset of 294,088 records. For the model development phase, this final dataset was randomly partitioned into three subsets: 80% allocated for training, 10% for validation, and the remaining 10% for testing.

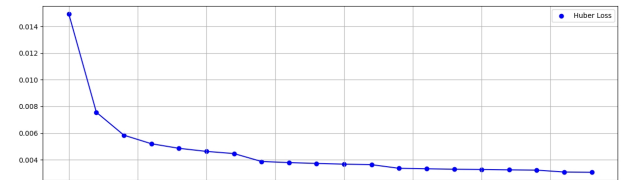
The model training was performed using the **Adam optimizer**, an adaptive learning rate algorithm, initialized with a learning rate of 10^{-3} . [4b]

A critical component of our approach is the choice of the loss function and, given the nature of seismic data, which often contains outliers and extreme ground motion values, we employed the **Huber Loss** [4a] instead of the standard Mean Squared Error (MSE). The Huber Loss is less sensitive to outliers, providing a robust regression metric defined as:

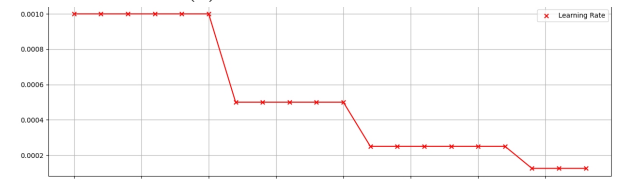
$$L_{\delta}(y, f(x)) = \begin{cases} \frac{1}{2}(y - f(x))^2 & \text{for } |y - f(x)| \leq \delta \\ \delta(|y - f(x)| - \frac{1}{2}\delta) & \text{otherwise} \end{cases}$$

where δ is the transition parameter we set at 0.5.

To ensure convergence and prevent overfitting, we implemented a learning rate scheduler (**ReduceLROnPlateau**) that reduces the learning rate by a factor of 0.1 if the validation loss does not improve for a defined patience period.



(a) Huber Loss Evolution



(b) Learning Rate Schedule

Figure 4: **Training Dynamics.** (a) Evolution of the Huber Loss across training epochs, showing convergence without overfitting. (b) Dynamic adjustment of the learning rate via the *ReduceLROnPlateau* scheduler to fine-tune weights.

2.2 Features

The Deep Neural Network is trained using a set of 18 input features, carefully selected to represent the seismic source, the wave propagation path, the local site conditions, and the reference motion recorded at the borehole. The features are derived from two primary sources: the KiK-net Flatfile (Loviknes et al., 2025[3]) and the Site Database of Strong-Motion Stations (Zhu et al., 2020[4]). The inputs are categorized into three main categories:

- Source & Path Parameters: characteristics of the seismic event and distance.
- Site Characteristics: geotechnical and topographical characteristics of the site.
- Borehole Motion Parameters : characteristics of the deep-depth motion, used as "forcing" input for the network.

The table [1] summarizes the 18 input variables used to train the Deep Neural Network, categorized by their physical meaning. Data sources are the KiK-net Flatfile [3] and the Site Database [4].

While the KiK-net database provides detailed borehole stratigraphies, directly encoding variable-length soil profiles into a neural network presents challenges regarding dimensionality and sparsity. Instead of using sequence-based architectures (e.g., RNNs), which might introduce overfitting given the dataset size, we opted to condense the stratigraphic information into robust scalar proxies (such as V_{S30} , $V_{S,bedrock}$, and fundamental frequency).

This approach preserves the essential physical characteristics of the site while maintaining a fixed-size input vector suitable for a stable Deep Neural Network regression.

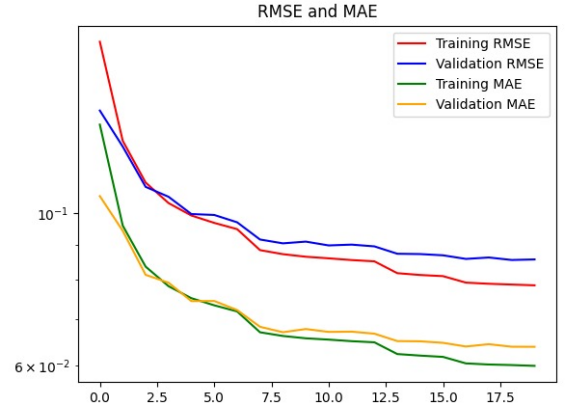
2.3 Confront & Validation

To rigorously validate the performance of the proposed DNN, we conducted a comparative analysis against two established ensemble learning algorithms: **Random Forest** (RF) and **XGBoost** (Gradient Boosting). These models served as strong baselines to demonstrate the superior capability of the deep learning approach in capturing the complexities of site amplification phenomena.[5]

Table 2: Results: Performance Metrics

Model	R^2	RMSE	MAE
DNN	0.889	0.007	0.064
XGBoost	0.813	0.112	0.085
Random Forest	0.756	0.131	0.098

Figure 5: Performance Metrics in semilogarithmic scale.



In the *figure 5* we can see consistent minimization of RMSE and MAE on training and validation set throughout the model's training process. The final values (RMSE=0.007, MAE=0.064) demonstrate the DNN's precision compared to baseline models. Blue and red trendmarks are referred to RMSE of Validation and Training set respectively; yellow and green ones are referred to MAE of Validation and Training set respectively.

The observed performance gap between the DNN and the ensemble methods (XGBoost, Random Forest) can be attributed to their fundamental structural differences. While tree-based algorithms rely on a rigid partitioning of the feature space—which often limits their ability to extrapolate outcomes for high-intensity events outside the training distribution—the DNN utilizes continuous activation functions (ReLU) to approximate the smooth, non-linear functions typical of physical wave propagation. Furthermore, the strategic implementation of the Huber Loss function proved critical for the model's stability: by linearly penalizing extreme residuals, it effectively desensitized the learning process to sporadic outliers inherent in large-scale seismic catalogs. In contrast, the standard MSE minimization likely forced the ensemble models to over-adjust to these outliers, resulting in significantly higher error metrics ($RMSE > 0.11$).

and reduced generalization capabilities.

3 Results & Discussion

To evaluate the predictive capability of the machine learning models on the site amplification function, the prediction target was defined as the base-10 logarithm of the ratio between surface and borehole Peak Ground Acceleration:

$$Y = \log_{10} \left(\frac{PGA_{surf}}{PGA_{bore}} \right) \quad (1)$$

The adoption of a logarithmic scale represents a standard practice in engineering seismology, as amplification factors typically follow a log-normal distribution and the transformation is fundamental to stabilize the variance of residuals and reduce the skewness of the distribution, allowing regression models to effectively handle dynamics spanning multiple orders of magnitude and to treat multiplicative errors as additive components.

The analysis of the metrics [2] obtained on the test set reveals a marked superiority of the **Deep Neural Network (DNN)** approach compared to the decision-tree-based algorithms. The DNN achieved a coefficient of determination $R^2 \approx 0.89$, demonstrating the ability to explain nearly 90% of the observed variance in the amplification ratio. Conversely, the ensemble models exhibited lower performance: **XGBoost** reached an $R^2 \approx 0.81$, while **Random Forest** showed the greatest difficulty in generalizing on the test data, yielding an $R^2 \approx 0.76$.

The residual analysis [5] confirms the robustness of the neural network. The **Mean Absolute Error (MAE)** of the DNN is 0.073, significantly lower than that of XGBoost (0.085) and Random Forest (0.098). A particularly significant gap is observed in the **Root Mean Squared Error (RMSE)**: the remarkably low value obtained by the DNN (0.007) suggests that the combination of the Huber Loss function with the Adam optimizer allowed the model to effectively minimize the impact of outliers. This led to a convergence towards a more stable solution compared to the standard mean squared error minimization, which appeared to penalize the tree-based models more severely ($RMSE > 0.11$).

3.1 Physical Interpretation

The success of the DNN suggests that the relationship between input parameters (source, path, and site characteristics) and ground motion amplification is dominated by strong non-linearities. The multi-layer architecture maps these non-linearities with greater precision than the rigid feature space partitioning performed by decision trees.

Notably, the inclusion of derived physical parameters, such as Δ_{dur} (defined as the difference between the significant durations D_{5-95} and D_{5-75}), provided the model with crucial information regarding the seismic signal coda and late-phase energy release. This feature likely enhanced the prediction of local resonance phenomena and wave trapping effects in superficial layers, which are often associated with prolonged ground motion duration.

The model’s ability to minimize residuals across a vast dynamic range implies that the network has successfully encoded the hysteretic behavior of soil deposits directly from the data. Specifically, the DNN implicitly captures the physical saturation of the amplification factor: as the input energy ($PGA_{borehole}$) increases, the model correctly predicts a non-linear damping response rather than a linear increase. The differential duration parameter (Δ_{dur}) validates the physical consistency of the learned features and effectively acts as a proxy for stratigraphic complexity and scattering; a larger Δ_{dur} indicates that seismic energy is being trapped within superficial layers due to multiple reflections, a mechanism that is physically correlated with prolonged ground shaking and enhanced local resonance phenomena.

4 Conclusions & Developments

This study demonstrated the efficacy of Deep Learning approaches in modeling the non-linear site amplification function, highlighting a significant performance gap between Deep Neural Networks (DNN) and traditional ensemble methods like Random Forest. The inclusion of physics-based features, such as the differential significant duration (Δ_{dur}), proved crucial in capturing the complex dynamics of energy accumulation.

However, some limitations must be acknowledged to properly contextualize these findings.

1. First, the model relies on a dataset derived exclusively from the Japanese KiK-net network. Consequently, the learned representations inevitably reflect specific regional geological settings and might suffer from limited generalization capabilities when applied to different tectonic environments (e.g., Europe or California) without domain adaptation strategies.
2. Second, despite their predictive power, DNNs remain "black-box" models. While they effectively map the input-output relationship, they do not provide an explicit analytical form of the amplification law, limiting the direct physical interpretability of the results compared to empirical ground motion models.

To overcome these limitations, future research should focus on two main technological advancements:

- **Physics-Informed Neural Networks (PINNs):** Instead of purely data-driven training, the loss function could be regularized by physical constraints (e.g., the wave equation or boundary conditions). This hybrid approach would ensure that the model's predictions are not only statistically accurate but also physically consistent, even in data-sparse regions.
- **Waveform-based Deep Learning:** The current approach relies on extracted scalar features (PGA, I_a). A significant leap forward would be the implementation of Convolutional Neural Networks (CNNs) or Recurrent Neural Networks (LSTMs) capable of processing raw acceleration time-series or spectrograms directly. This would allow the model to automatically extract latent features related to the frequency content and phase of the signal, potentially unveiling complex site-response patterns that manual feature engineering might overlook.

In conclusion, while the proposed DNN architecture serves as a robust baseline for site response prediction, the integration of physical laws into the learning process represents the necessary next step towards a generalized and interpretable AI-based seismic risk assessment tool.

References

- [1] National Research Institute for Earth Science and Disaster Resilience . NIED k-net, kik-net, national research institute for earth science and disaster resilience, 2019. URL https://www.kyoshin.bosai.go.jp/kyoshin/docs/overview_kyoshin_en.shtml.
- [2] Google. Gemini. Generative AI model, 2024. URL <https://gemini.google.com/>. Image generated in response to the prompt: "Inserisci qui il tuo prompt in inglese".
- [3] K. Loviknes, S. von Specht, H. Lilienkamp, A. Händel, and F. Cotton. KiK-net and K-NET flatfile with automatically processed ground motions recorded between 1996 and 2024, 2025. URL <https://doi.org/10.5880/GFZ.LKUT.2025.001>.
- [4] C. Zhu, G. Weatherill, F. Cotton, M. Pilz, D. Y. Kwak, and H. Kawase. An Open-Source Site Database of Strong-Motion Stations in Japan: K-NET and KiK-net, 2020. URL <https://doi.org/10.5880/GFZ.2.1.2020.006>.

Table 1: List of Input Features.

Variable Name	Source	Description
1. Source & Path Parameters		
fnet_MT_Magnitude(Mw)	Flatfile	Moment Magnitude (M_w) derived from the F-net catalog.
fnet_MT_Depth(km)	Flatfile	Event depth derived from the F-net catalog (km).
Rhypo	Flatfile	Hypocentral distance, representing the direct path length from source to site.
2. Site Characteristics		
VS, sb	Site DB	Average S-wave velocity from the ground surface to the borehole sensor depth.
VS30	Site DB	Time-averaged shear-wave velocity in the upper 30 m, inferred from topographic slope.
Slope [Horn]	Site DB	Local ground slope derived from digital terrain models using Horn's algorithm.
TRI	Site DB	Terrain Ruggedness Index: morphometric measure of elevation difference.
Borehole_depth	Flatfile	Depth from the surface to the installed borehole sensor.
3. Borehole Motion Parameters (Input Forcing)		
PGA_rotD50_B	Flatfile	Peak Ground Acceleration (RotD50) recorded at the borehole sensor.
PGV_rotD50_B	Flatfile	Peak Ground Velocity (RotD50) of the processed waveform at the borehole.
AriasIntensity	Flatfile	Arias Intensity (I_A), calculated as the time-integral of squared acceleration, we took the component for the EW and NS doing the norm 2.
$\Delta Dur = Dur_{95} - Dur_{75}$	Flatfile	Represents the time interval during which the Arias intensity increases from 75% to 95% of its total value, .
fc0_B	Flatfile	Low-cut corner frequency used in filtering (max 0.5 Hz).
B0.020	Flatfile	5% Damped Spectral Acceleration (SA) at borehole for period $T = 0.020$ s.
B0.050	Flatfile	5% Damped Spectral Acceleration (SA) at borehole for period $T = 0.050$ s.
B0.200	Flatfile	5% Damped Spectral Acceleration (SA) at borehole for period $T = 0.200$ s.
B0.500	Flatfile	5% Damped Spectral Acceleration (SA) at borehole for period $T = 0.500$ s.
B1.000	Flatfile	5% Damped Spectral Acceleration (SA) at borehole for period $T = 1.000$ s.